EXPERIMENT 14

AIM:

To understand and implement feature scaling techniques on a given dataset.

CODE:

```
import numpy as np
import pandas as pd

# Upload the dataset (Google Colab)
from google.colab import files
uploaded = files.upload()

# Load the dataset
df = pd.read_csv("feature_scaling_dataset.csv")
print(df)

# Handle missing values in Country
df["Country"] = df["Country"].fillna(df["Country"].mode()[0])
print(df)

# Separate features and label
features = df.iloc[:, :-1].values
label = df.iloc[:, -1].values
print(features)

# Import SimpleImputer
```

```
from sklearn.impute import SimpleImputer

# Create imputers
age_imputer = SimpleImputer(strategy="mean")
salary_imputer = SimpleImputer(strategy="mean")

# Fit the imputers
age_imputer.fit(features[:, [1]])
salary_imputer.fit(features[:, [2]])

# Replace missing values
features[:, [1]] = age_imputer.transform(features[:, [1]])
features[:, [2]] = salary_imputer.transform(features[:, [2]])
print(features)

# Apply One-Hot Encoding
from sklearn.preprocessing import OneHotEncoder

oh = OneHotEncoder(sparse_output=False)

Country = oh.fit_transform(features[:, [0]])
print(Country)

# Combine encoded and numerical features
final_set = np.concatenate(
```

```
    (Country, features[:, [1, 2]]),
    axis=1
)
print(final_set)

# Apply StandardScaler
from sklearn.preprocessing import StandardScaler

sc = StandardScaler()
sc.fit(final_set)

feat_standard_scaler = sc.transform(final_set)
print(feat_standard_scaler)

# Apply MinMaxScaler
from sklearn.preprocessing import MinMaxScaler

mms = MinMaxScaler(feature_range=(0, 1))
mms.fit(final_set)

feat_minmax_scaler = mms.transform(final_set)
print(feat_minmax_scaler)
```

OUTPUT:

1. Original Dataset

```
   Country   Age   Salary Purchased
0   France  42.0  68000.0        No
1    Spain  29.0  51000.0       Yes
2  Germany  31.0  57000.0        No
3    Spain  36.0  64000.0       Yes
4  Germany   NaN      NaN        No
5   France  34.0  61000.0       Yes
6    Spain   NaN  55000.0        No
7   France  46.0  82000.0       Yes
8  Germany  51.0  76000.0        No
9   France  39.0  70000.0       Yes
```

2. Features After Missing Values Are Filled

```
array([['France', 42.0, 68000.0],
       ['Spain', 29.0, 51000.0],
       ['Germany', 31.0, 57000.0],
       ['Spain', 36.0, 64000.0],
       ['Germany', 38.5, 64888.88888888889],
       ['France', 34.0, 61000.0],
       ['Spain', 38.5, 55000.0],
       ['France', 46.0, 82000.0],
       ['Germany', 51.0, 76000.0],
       ['France', 39.0, 70000.0]], dtype=object)
```

3. One-Hot Encoding

```
array([[1., 0., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [1., 0., 0.],
       [0., 0., 1.],
       [1., 0., 0.],
       [0., 1., 0.],
       [1., 0., 0.]])
```

4. Final Dataset Before Scaling

```
array([[1.0, 0.0, 0.0, 42.0, 68000.0],
       [0.0, 0.0, 1.0, 29.0, 51000.0],
       [0.0, 1.0, 0.0, 31.0, 57000.0],
       [0.0, 0.0, 1.0, 36.0, 64000.0],
       [0.0, 1.0, 0.0, 38.5, 64888.88888888889],
       [1.0, 0.0, 0.0, 34.0, 61000.0],
       [0.0, 0.0, 1.0, 38.5, 55000.0],
       [1.0, 0.0, 0.0, 46.0, 82000.0],
       [0.0, 1.0, 0.0, 51.0, 76000.0],
       [1.0, 0.0, 0.0, 39.0, 70000.0]], dtype=object)
```

5. StandardScaler Output

```
array([[ 1.22474487, -0.65465367, -0.65465367,  0.5547873 ,  0.34337869],
       [-0.81649658, -0.65465367,  1.52752523, -1.50585123, -1.53294056],
       [-0.81649658,  1.52752523, -0.65465367, -1.18882992, -0.87071024],
       [-0.81649658, -0.65465367,  1.52752523, -0.39627664, -0.0981082 ],
       [-0.81649658,  1.52752523, -0.65465367,  0.        ,  0.        ],
       [ 1.22474487, -0.65465367, -0.65465367, -0.71329795, -0.42922336],
       [-0.81649658, -0.65465367,  1.52752523,  0.        , -1.09145368],
       [ 1.22474487, -0.65465367, -0.65465367,  1.18882992,  1.88858277],
       [-0.81649658,  1.52752523, -0.65465367,  1.9813832 ,  1.22635245],
       [ 1.22474487, -0.65465367, -0.65465367,  0.07925533,  0.56412213]])
```

6. MinMaxScaler Output

```
array([[1.        , 0.        , 0.        , 0.59090909, 0.5483871 ],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 1.        , 0.        , 0.09090909, 0.19354839],
       [0.        , 0.        , 1.        , 0.31818182, 0.41935484],
       [0.        , 1.        , 0.        , 0.43181818, 0.44802867],
       [1.        , 0.        , 0.        , 0.22727273, 0.32258065],
       [0.        , 0.        , 1.        , 0.43181818, 0.12903226],
       [1.        , 0.        , 0.        , 0.77272727, 1.        ],
       [0.        , 1.        , 0.        , 1.        , 0.80645161],
       [1.        , 0.        , 0.        , 0.45454545, 0.61290323]])
```

RESULT:

The dataset was successfully scaled using StandardScaler and MinMaxScaler.